# NARENDARAN S G

**Final-Year ECE Student • ASIC Physical Design (Back-End) • RTL Design & Verification (Front-End) • DFT**

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## EDUCATION

**Easwari Engineering College (SRM Group)**, Chennai, India — **Sep 2023 – Jul 2027**
*Bachelor of Engineering (B.E.) in Electronics and Communication Engineering (ECE)* — **CGPA: 9.30**

**Chettinad Vidyashram**, Chennai, India — *Higher Secondary School (Science Stream: PCM, Biotechnology)* — **Class XII, May 2023 | Grade: 96%**

## EDA TOOLCHAINS & TECHNICAL SKILLS

**Commercial EDA Suite:** Synopsys IC Compiler II (`icc2_shell`), Synopsys Design Compiler (`dc_shell`), Synopsys Design Vision, Synopsys IC Compiler, Synopsys VCS, Synopsys PrimeTime, Cadence Innovus, Cadence Genus, Cadence Modus.

**Open-Source EDA & PDK:** OpenLane, OpenROAD, Yosys, Verilator, Magic, KLayout, SkyWater 130nm PDK (`sky130_fd_sc_hd`).

**HDLs & Verification:** Verilog HDL, SystemVerilog HVL, UVM Fundamentals, Coverage-Driven Verification (CDV), OOP Testbenches, Switch-Level CMOS.

**Physical Design & DFT:** Floorplanning, Flyline Macro Placement, Power Delivery Network (PDN) Mesh, Congestion Mitigation, Multi-Clock CTS & NDR (`dwds`), STA Timing Closure, DRC/LVS & Route ECO, Scan-Chain Insertion, PCB Design.

## ASIC PHYSICAL DESIGN, VERIFICATION & DFT PROJECTS

**8-Bit Asynchronous FIFO Physical Design & Routing Optimization — Synopsys IC Compiler II** — Apr 2026 – May 2026

- Implemented an end-to-end layout across a 2,065.428 µm² core area (63.14% initial → 68.41% final utilization, 1 µm core offset, 305 final cells) with 24 track-aligned M5 ports (West inputs, East outputs).
- Structured an M7/M8 top mesh (0.6 µm width, 18 µm pitch), M2 vertical straps (0.18 µm width, 6 µm pitch), and 0.06 µm M1 rails with 97 DCAP and 2 TIEH cells, fixing 2 via spacing violations for 0 PG DRC and 0% placement congestion.
- Balanced dual asynchronous clocks (`wclk`: 1.50 ns period, 80 sinks, 0.51 ns latency; `rclk`: 1.50 ns period, 24 sinks, 0.45 ns latency) with 34 buffers/18 inverters on M4–M5 (`dwds` NDR) achieving 0 ns skew within each clock domain, 0 DRC/Shorts/Opens, +0.16 ns Setup WNS, and 0 ns Hold TNS.

**Verilog Half Adder: Multi-Abstraction Implementation & Functional Verification — Synopsys VCS** — May 2026

- Mapped a combinational half adder across Dataflow (`assign`), Behavioral (`always @(*)`), Gate-Level Structural, and Switch-Level CMOS (`pmos`/`nmos` networks wired to `supply1 VDD`/`supply0 VSS` via `nand_sw`, `nor_sw`, `inverter_sw`), verified simultaneously via a SystemVerilog testbench (`tb_top`) on Synopsys VCS (github.com/narendaransg/half-adders-at-different-forms).

**GPRS Block Physical Design (`gprs_top`) — Synopsys Design Compiler & IC Compiler II** — Apr 2026

- Executed RTL-to-signoff physical design on a 10-metal-layer block with 27,811 initial cells (33,286 final cells), 16 hard SRAM macros (4 families), 103 track-aligned ports (M5 West / M6 South), and a 391,605.407 µm² core area.
- Oriented 16 macros using data-flow flylines (>10 iterations, 16 µm channel spacing, soft blockages) and built an M8/M7/M2 power mesh down to 0.06 µm M1 rails; added 179 missing PG vias to achieve 0 PG DRC and 0 PG shorts.
- Placed 8,600 DCAP, 76 TIEH, and 121 TIEL cells at 62.39% density (0.12% horizontal congestion); synthesized `mclk` (4 ns period, 2,856 sinks, 65 buffers/inverters, M4–M5 `dwds` NDR) with 0.14 ns global skew and resolved 34 route shorts via `route_eco` (66.25% final utilization, 0 ns Setup WNS/TNS).

**Verilog Multi-Master Bus Arbiter — ChipThink VLSI Hackathon Finalist (Maven Silicon & IEEE)** — Apr 2026

- Designed and verified an offline Verilog Multi-Master Bus Arbiter for CPU, DMA, and Peripheral contention during the finals at Christ College of Engineering (edaplayground.com/x/JiD3).
- Engineered priority-based CPU arbitration with an 8-clock-cycle fairness counter that forces priority rotation for waiting DMA/Peripheral masters to prevent starvation, complete with hardware debug state outputs and waveform verification.

**Serial-Parallel Multiplier (SPM) RTL-to-GDSII (OpenLane/SKY130) & PLL RTL-to-Routing (Synopsys)** — Nov 2025 – Apr 2026

- Executed a 70+ step automated RTL-to-GDSII flow for an SPM using Yosys, Verilator, OpenROAD, Magic, and KLayout in the SkyWater 130nm PDK (`sky130_fd_sc_hd`), and completed RTL-to-routing with pre-layout simulation for a Phase-Locked Loop (PLL) in Synopsys Design Vision & IC Compiler.

**ASIC Core Physical Implementation (Cadence Innovus) & Memory Block DFT (Cadence Genus/Modus)** — Nov 2025

- **Cadence Innovus PD:** Placed standard cells at 53.189% density (Pre-CTS WNS: 0.846 ns), synthesized clock tree with 0.011 ns skew (Post-CTS WNS improved to 1.377 ns via `optDesign`), and routed with 0.00% routing overflow.
- **Cadence Genus & Modus DFT:** Performed scan-chain insertion, stitched scan flops, validated shift/capture modes, cleared DFT rule checks, and generated timing-aware synthesis PPA reports for a memory block.

**4-Bit Binary Fast Multiplier in IC Level & 8-Bit Verilog Barrel Shifter Synthesis** — Oct 2024 – Jan 2025 / Nov 2025

- Assembled a breadboard 4-Bit Binary Fast Multiplier (15×15 = 225 / `11100001`) using discrete AND gate arrays and 4-bit full adders (Oct 2024 – Jan 2025) and analyzed multi-stage multiplexer synthesis of an 8-bit Verilog Logical Shifter (Nov 2025).

## INDUSTRIAL INTERNSHIP

**Nokia** — *Student Intern (Factory Floor & Process Training — Embedded Systems & PCB Design)* — **May 2025 – Jun 2025 | Sriperumbudur**

Gained exposure to live telecommunications hardware factory floor operations, PCB processes, and industrial operational efficiency.

## CERTIFICATIONS, WORKSHOPS & LEADERSHIP

- **Physical Design – Back End Certification:** Centre for Chip Design, Easwari Engineering College & VLSIMINDS, Bangalore (21 Apr 2026).
- **VLSI Design Verification – Front End Certification:** Centre for Chip Design, Easwari Engineering College & VLSIMINDS, Bangalore (04 Feb 2026).
- **Two-Day Workshop on Physical Design using Cadence Tools:** Dept. of ECE & Centre for Chip Design, Easwari Engineering College (07–08 Nov 2025).
- **Hackathons & Affiliations:** Finalist, ChipThink VLSI Hackathon (Maven Silicon & IEEE, Apr 2026); Participant, CHIPCRAFT 2.0 (Kalasalingam Academy, Apr 2026); Student Member, IETE (Jul 2024 – Jul 2026) & ISTE (Oct 2024 – Jul 2026).